

MoLA: Molecular multimodal layerwise adaptive network for molecular property prediction

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ABSTRACT

Molecular property prediction is pivotal in drug discovery and materials science, where leveraging multimodal molecular representations has become a central strategy for enhancing predictive performance. However, most existing multimodal fusion frameworks adopt shallow or rigid fusion mechanisms, implicitly assuming that simple aggregation of heterogeneous features is sufficient to capture complex biochemical characteristics. This overlooks a key challenge: the rationality and adaptiveness of modality integration remain largely underexplored, despite different modalities contributing unequally to diverse molecular properties. Naive fusion may introduce redundancy, noise, or even obscure essential cross-modal relationships. To address these limitations, we propose **Molecular Multimodal Layerwise Adaptive Network (MoLA)**, a novel multilayer neural architecture designed for molecular property prediction. At its core, MoLA employs a Cross-layer Attention Fusion mechanism to adaptively construct a Multimodal Correlation Map, enabling selective and nuanced information exchange both across modalities and network depths. Notably, MoLA further integrates a dendritic neuron module within the fingerprint branch, enhancing its capacity to capture complex, context-dependent feature interactions inspired by biological neural computation. Extensive experiments on multiple benchmarks from the MoleculeNet suite demonstrate that MoLA achieves state-of-the-art performance across a variety of molecular property prediction tasks, underscoring the importance of rational and adaptive multimodal fusion in computational chemistry. The code implementation is available at the provided repository: <https://github.com/Rirock/MoLA>.

1. Introduction

Molecular properties prediction is crucial in various scientific fields such as drug discovery, environmental chemistry, and material sciences [1,2]. Accurately predicting molecular characteristics can significantly accelerate compound screening, reduce experimental costs, and facilitate the development of effective therapeutics [3,4]. Traditional computational approaches, including Quantitative Structure-Activity Relationship (QSAR) models, often rely on manually designed molecular descriptors and single-modality inputs, limiting their predictive power and generalizability [5].

Recently, machine learning and deep learning methods have greatly enhanced molecular prediction tasks through the use of various molecular representations [6,7]. The Simplified Molecular Input Line Entry System (SMILES), a textual representation of molecules, effectively captures

detailed sequential information of atoms and bonds [8,9]. SMILES-based methods, particularly those employing Transformer architectures, excel in modeling sequential dependencies and extracting semantic chemical information but lack explicit spatial structural context [7]. Molecular fingerprints, such as Extended-Connectivity Fingerprints (ECFP), provide concise and efficient representations by summarizing molecular substructures and their occurrences, making them computationally appealing for large-scale analyses [10,11]. However, fingerprints abstract away detailed structural nuances, losing explicit relational and spatial information [12]. Graph-based representations explicitly encode spatial connectivity and relational details through molecular graphs, with Graph Neural Networks (GNNs) demonstrating strong capabilities in modeling molecular topology [13–15]. Despite their strengths in capturing local structural patterns, graph representations often overlook global chemical properties and detailed atom-level sequences [16].

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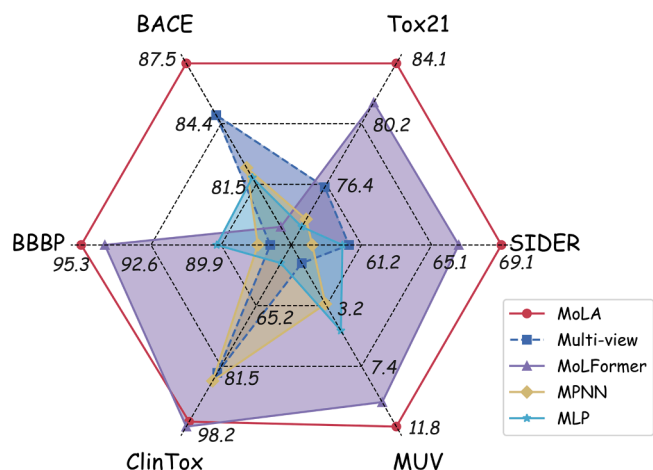


Fig. 1. Performance comparison of MoLA and other models, detailed data is shown in Table 3.

To overcome the limitations inherent to single-modal approaches, researchers have increasingly adopted multimodal frameworks that integrate multiple molecular representations [17]. Recent multimodal methods include the combination of graph and fingerprint representations using GNN and fully connected neural networks [12], or the integration of SMILES and graph representations leveraging both Transformer and GNN architectures [18]. These methods have shown improved predictive performance by exploiting complementary information across modalities. For instance, previous studies demonstrated that combining SMILES and graph representations could enhance both sequential and structural feature extraction, significantly improving prediction accuracy in various molecular tasks [19]. However, current multimodal approaches typically adopt shallow fusion techniques, where each modality is independently processed through distinct neural networks, and only their final representations are merged. Such methods fail to fully leverage the deep hierarchical interactions and rich complementary information across modalities, limiting their predictive capabilities [12,20,21].

To address these limitations, we present MoLA, a Molecular Multimodal Layerwise Adaptive Network designed for molecular property prediction. MoLA achieves hierarchical and adaptive fusion of heterogeneous molecular representations, such as molecular graphs, fingerprints, and SMILES, at both shallow and deep layers of the network. In contrast to conventional approaches that only merge outputs at the final stage, MoLA enables dynamic information exchange and selective integration between modalities throughout the entire feature extraction process. This design allows the model to capture fine-grained and task-relevant interactions across multiple levels of abstraction. In addition, MoLA incorporates a dendritic neuron module within the fingerprint processing pathway. Inspired by the compartmentalized computation in biological neurons, the dendritic neuron module decomposes feature transformation into localized synaptic and dendritic nonlinearities, followed by somatic aggregation. This mechanism enables the network to selectively enhance important molecular features, thereby improving both expressivity and interpretability [22].

Although MoLA is composed of established architectural components, its contribution lies in the principled coordination of cross-layer fusion and fingerprint-specific dendritic modeling within a unified framework. Unlike prior approaches that perform fusion at a single abstraction level or treat fingerprint encoders as generic MLPs, MoLA introduces an adaptive, layer-wise integration mechanism that allows different modalities and representation depths to interact dynamically (Fig. 1).

We evaluate MoLA on twelve benchmark datasets from MoleculeNet, covering both classification and regression tasks. Experimental results

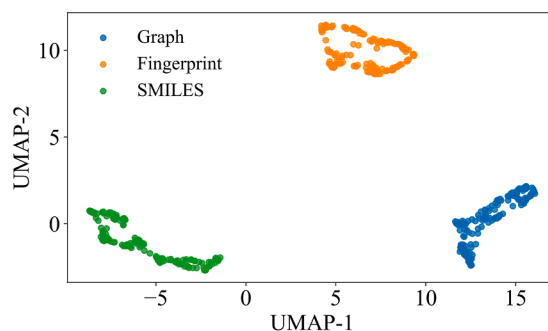


Fig. 2. UMAP projection of molecular fingerprints, SMILES, and graph-based representations. The clear separation among modalities demonstrates their inherent differences, suggesting that naive addition or concatenation is insufficient for capturing cross-modal relationships.

demonstrate that MoLA consistently outperforms state-of-the-art unimodal and multimodal baselines, achieving superior predictive performance across diverse chemical endpoints. Our contributions are as follows:

- 1) We design **MoLA**, a novel architecture enabling hierarchical, adaptive multimodal fusion across network layers.
- 2) We introduce a biologically inspired **Dendritic Neuron Module** that enhances the modeling capacity of molecular fingerprints.
- 3) We conduct extensive experiments on twelve public benchmarks, achieving new state-of-the-art results and demonstrating the effectiveness of our approach.

2. Related work

Molecular property prediction has evolved from traditional machine learning using handcrafted descriptors to deep learning approaches that automatically extract features from molecular fingerprints, SMILES strings, and molecular graphs [23–25]. Graph Neural Networks (GNNs) have demonstrated strong performance by directly modeling molecular structures [26]. However, single-modal methods remain limited, as no individual representation can fully capture the complexity of molecular properties [27].

To address these limitations, multimodal learning frameworks integrate information from multiple molecular representations [17]. Methods such as FP-GNN [12] and Multi-view [28] process fingerprints, graphs, and SMILES independently and then combine the extracted features for final prediction. Most current approaches implement fusion by concatenating or linearly combining features at the output stage. This practice raises several challenges. First, modality-specific abstraction may be required at different network depths. Second, interactions between modalities may evolve through the network and are not captured by late-stage fusion. Third, existing methods lack mechanisms to adaptively balance heterogeneous information in a task-aware manner (Fig. 2).

Recent work in other fields has demonstrated the effectiveness of deep attention-based fusion for capturing complex interactions among heterogeneous modalities [29,30]. In contrast, most multimodal molecular property prediction models still perform fusion at a single network layer and treat it as a post-processing step, limiting their ability to model cross-modal relationships [31].

In summary, current multimodal molecular property prediction approaches often place limited emphasis on hierarchical and adaptive integration across different molecular representations. To address this limitation, we propose MoLA, a novel network architecture that performs layer-wise adaptive fusion of heterogeneous molecular representations and incorporates a dendritic neuron module specifically tailored for fingerprint feature processing. By explicitly modeling adaptive interactions

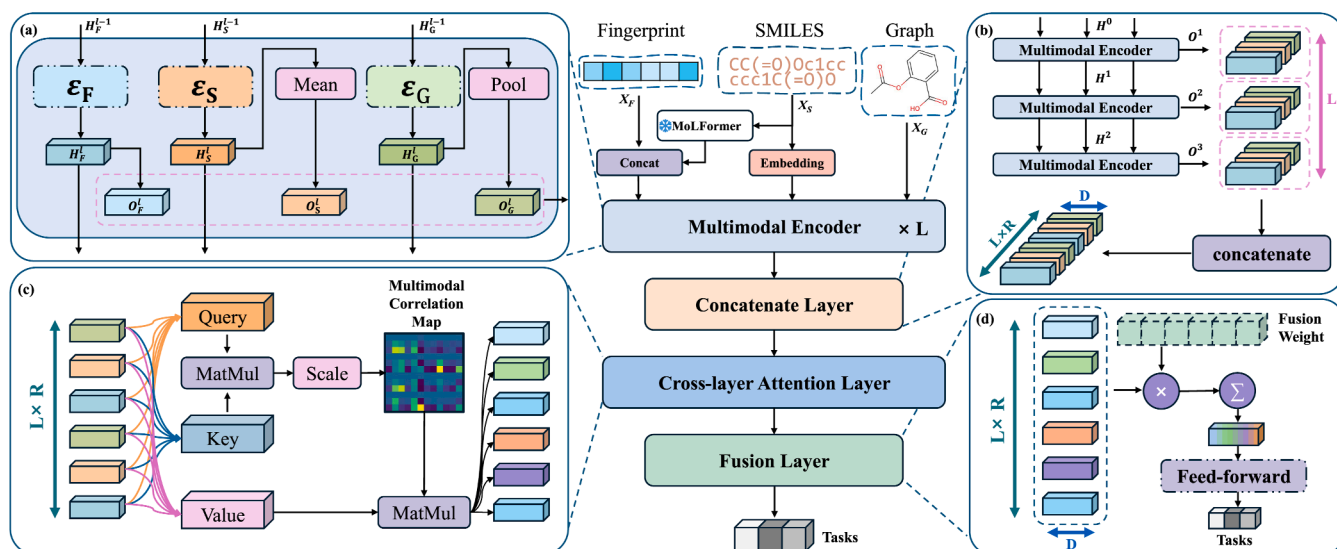


Fig. 3. Overall architecture of MoLA for molecular property prediction. (a) Multimodal Encoder separately encodes fingerprints, SMILES, and molecular graphs, with the dendritic neuron module applied to the fingerprint branch. (b) Concatenate Layer merges the representations from all modalities into a unified vector. (c) Cross-layer Attention Layer adaptively integrates multi-level and multi-modal features via self-attention. (d) Fusion Layer generates the final molecular representation for property prediction.

across modalities and network layers, MoLA is designed to capture nuanced and dynamic relationships that are difficult to represent with conventional fusion strategies. The dendritic neuron module further enables selective and non-linear feature interactions within high-dimensional fingerprint representations, leading to more rational, effective, and interpretable molecular property prediction.

3. Methodology

3.1. Problem definition

Molecular property prediction is the task of estimating chemical or biological properties of a molecule based on its computational representations. Formally, given a molecule represented by multimodal data $\mathbf{X}$, the objective is to learn a predictive function F such that $y = F(\mathbf{X})$, where y denotes the target property and $\mathbf{X}$ encompasses various molecular representations, including fingerprints, SMILES strings, and graph-based structures.

3.2. Structure overview

MoLA is a multimodal framework designed for molecular property prediction and consists of four main components: the Multimodal Encoder, Concatenate Layer, Cross-layer Attention Layer, and Fusion Layer. The Multimodal Encoder processes different molecular representations through separate branches, with the dendritic neuron module applied exclusively to the fingerprint branch to enhance its nonlinear feature extraction. The subsequent layers perform hierarchical integration and adaptive interaction among modalities. Fig. 3 provides an overview of the architecture and illustrates the sequential flow of information through each module.

3.3. Preprocessing and feature representation

We leverage three primary molecular data modalities: molecular fingerprints $\mathbf{X}_F$, SMILES $\mathbf{X}_S$, and molecular graphs $\mathbf{X}_G$. Each modality is preprocessed and embedded as follows:

Fingerprint Embedding. To integrate both expert-crafted and data-driven features, we concatenate traditional molecular fingerprints and dense

representations from a pretrained MolFormer model applied to SMILES. The resulting fingerprint embedding is given by:

$$\mathbf{H}_F^0 = \text{Concat}(\mathbf{X}_F, \text{MolFormer}(\mathbf{X}_S)) \in \mathbb{R}^{d_F + d_M} \quad (1)$$

where $\mathbf{X}_F \in \mathbb{R}^{d_F}$ denotes the binary fingerprint vector, and d_M is the dimension of the MolFormer embedding.

SMILES Embedding. For SMILES strings, each sequence $\mathbf{X}_S = \{\mathbf{x}_{S,1}, \mathbf{x}_{S,2}, \dots, \mathbf{x}_{S,t}\}$ is tokenized and mapped to a dense representation:

$$\mathbf{H}_S^0 = \text{Embedding}(\mathbf{X}_S) \in \mathbb{R}^{t \times d_S} \quad (2)$$

where t denotes the sequence length and d_S is the embedding size.

Graph Embedding. Molecular graphs are defined as $\mathbf{X}_G = (V, E)$, where $V \in \mathbb{R}^{v \times d_G}$ denotes node features. The initial graph embedding is set as:

$$\mathbf{H}_G^0 = V \quad (3)$$

where v is the number of nodes and d_G is the node feature dimension.

These initial representations $\mathbf{H}_F^0$, $\mathbf{H}_S^0$, and $\mathbf{H}_G^0$ are provided to the first layer of the multimodal encoder, enabling comprehensive integration of structural and semantic molecular information.

3.4. Multimodal encoder

The Multimodal Encoder is designed to integrate heterogeneous molecular information from multiple representation modalities across several network layers. For each molecule, we consider R input types (here $R = 3$: molecular fingerprints, SMILES strings, and molecular graphs), denoted as $\mathbf{X}_r$ for $r \in \{F, S, G\}$. The encoder consists of L sequential layers, where each layer l updates modality-specific hidden states and generates fixed-length outputs for subsequent fusion.

Formally, at each layer $l \in \{1, \dots, L\}$, the modality-specific encoders are defined as follows:

$$\mathbf{H}_r^l = \mathcal{E}_r(\mathbf{H}_r^{l-1}), \quad \mathbf{O}_r^l = \mathcal{P}_r(\mathbf{H}_r^l), \quad r \in F, S, G \quad (4)$$

where $\mathcal{E}_r$ is the dedicated encoder for modality r , and $\mathcal{P}_r$ is the modality-specific pooling or projection operator to produce a fixed-length output $\mathbf{O}_r^l \in \mathbb{R}^d$ at each layer. All modality-specific outputs are projected to a unified dimension d to facilitate multimodal integration.

Specifically, we instantiate the modality-specific encoders as follows:

- **Fingerprint Encoder ($\mathcal{E}_F$):** We adopt a Dendritic Neuron Module (DNM, $\mathcal{E}_F = \text{DNM}$) to enhance the nonlinear modeling capacity and capture high-order interactions among fingerprint features:

$$\mathbf{H}_F^l = \mathcal{E}_F(\mathbf{H}_F^{l-1}), \quad \mathbf{O}_F^l = \mathbf{H}_F^l \quad (5)$$

The initial input $\mathbf{H}_F^0$ concatenates raw fingerprints and MolFormer pretrained representations.

- **SMILES Encoder ($\mathcal{E}_S$):** To model the sequential structure of SMILES, we employ a Transformer encoder ($\mathcal{E}_S = \text{Transformer}$), followed by mean pooling:

$$\mathbf{H}_S^l = \mathcal{E}_S(\mathbf{H}_S^{l-1}), \quad \mathbf{O}_S^l = \text{MeanPool}(\mathbf{H}_S^l) \quad (6)$$

- **Graph Encoder ($\mathcal{E}_G$):** For molecular graphs, a Graph Isomorphism Network (GIN, $\mathcal{E}_G = \text{GIN}$) is used to encode topological information, with pooling to obtain graph-level features:

$$\mathbf{H}_G^l = \mathcal{E}_G(\mathbf{H}_G^{l-1}, \mathbf{B}), \quad \mathbf{O}_G^l = \text{Pool}(\mathbf{H}_G^l) \quad (7)$$

where $\mathbf{B}$ denotes the adjacency matrix.

In summary, we denote each modality encoder as $\mathcal{E}_r$ with corresponding outputs $\mathbf{O}_r^l$, enabling flexible and hierarchical multimodal representation. The specific choices for each $\mathcal{E}_r$ in this work are: DNM for fingerprints, Transformer for SMILES, and GIN for graphs.

3.5. Concatenate layer

The Concatenate Layer is designed to aggregate outputs from all molecular representations and all encoder layers, thereby enabling comprehensive cross-representation and cross-layer interactions. For each layer l and representation r , the output vector is denoted as $\mathbf{O}_r^l \in \mathbb{R}^d$. The concatenated tensor $\mathbf{Z}$ is constructed as follows:

$$\mathbf{Z} = \text{Concat}(\mathbf{O}_r^l \mid l = 1, \dots, L; r \in F, S, G) \quad (8)$$

where $\mathbf{Z} \in \mathbb{R}^{L \times R \times d}$ collects all outputs across layers and representations into a single tensor. For the subsequent attention and fusion modules, we flatten the first two dimensions:

$$\mathbf{Z}_{\text{flat}} = \text{Reshape}(\mathbf{Z}, [L \cdot R, d]) \quad (9)$$

This approach ensures that the network preserves the full spectrum of hierarchical and multimodal information for adaptive aggregation in later stages.

3.6. Cross-layer attention layer

To adaptively integrate information across all encoder layers and molecular modalities, we employ a multi-head self-attention block over the concatenated token sequence formed by stacking layer-wise representations from every modality. Let L denote the number of encoder layers per modality and R the number of modalities. After concatenation, we obtain $\mathbf{Z}_{\text{flat}} \in \mathbb{R}^{LR \times d}$, where each of the LR tokens corresponds to a specific (layer, modality) pair and d is the hidden dimension. The attention module enables every token to selectively aggregate context from all other tokens, thereby capturing hierarchical (cross-layer) and cross-modal dependencies that are crucial for accurate molecular property prediction.

For each attention head $h \in \{1, \dots, H\}$, we project $\mathbf{Z}_{\text{flat}} \in \mathbb{R}^{LR \times d}$ to queries, keys, and values:

$$\mathbf{Q}^{(h)} = \mathbf{Z}_{\text{flat}} \mathbf{W}_Q^{(h)}, \quad \mathbf{K}^{(h)} = \mathbf{Z}_{\text{flat}} \mathbf{W}_K^{(h)}, \quad \mathbf{V}^{(h)} = \mathbf{Z}_{\text{flat}} \mathbf{W}_V^{(h)}, \quad (10)$$

where $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d \times d_a}$ and d_a is the per-head dimension (so that $Hd_a = d$). Scaled dot-product attention computes pairwise scores

$$\mathbf{A}_{ij}^{(h)} = \frac{\langle \mathbf{Q}_i^{(h)}, \mathbf{K}_j^{(h)} \rangle}{\sqrt{d_a}}, \quad (11)$$

which are normalized over j to obtain attention weights

$$\alpha_{ij}^{(h)} = \frac{\exp(\mathbf{A}_{ij}^{(h)})}{\sum_{k=1}^{LR} \exp(\mathbf{A}_{ik}^{(h)})}. \quad (12)$$

Here i denotes the index of the query token, and j denotes the index of the key/value token, each corresponding to a specific (layer, modality) pair in $\mathbf{Z}_{\text{flat}}$. Each head then produces a context-aware representation via a weighted value aggregation

$$\tilde{\mathbf{Z}}_i^{(h)} = \sum_{j=1}^{LR} \alpha_{ij}^{(h)} \mathbf{V}_j^{(h)} \in \mathbb{R}^{d_a}, \quad (13)$$

and all heads are concatenated to yield the final attended sequence

$$\tilde{\mathbf{Z}}_i = \text{Concat}(\tilde{\mathbf{Z}}_i^{(1)}, \dots, \tilde{\mathbf{Z}}_i^{(H)}) \mathbf{W}^O, \quad \mathbf{W}^O \in \mathbb{R}^{(Hd_a) \times d}. \quad (14)$$

Equivalently, the multi-head self-attention operation can be written in the standard form as

$$\text{MultiHeadAttn}(\mathbf{Z}_{\text{flat}}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O, \quad (15)$$

where

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^\top}{\sqrt{d_a}}\right) \mathbf{V}^{(h)}, \quad \mathbf{W}^O \in \mathbb{R}^{(Hd_a) \times d}. \quad (16)$$

The concatenated multi-head outputs are linearly projected back to the d -dimensional space, yielding

$$\tilde{\mathbf{Z}} = \text{MultiHeadAttn}(\mathbf{Z}_{\text{flat}}) \in \mathbb{R}^{LR \times d}. \quad (17)$$

The learned attention weights provide interpretability by revealing which (layer, modality) tokens contribute most to each output token. The resulting context-enriched sequence $\tilde{\mathbf{Z}}$ is then forwarded to the Fusion Layer for subsequent integration and prediction.

Unlike conventional multimodal fusion strategies that operate on final-layer representations, the proposed cross-layer attention enables adaptive interaction across all encoder depths. This design allows MoLA to selectively exploit low-level structural cues or high-level semantic features depending on the prediction task, providing a more flexible and expressive fusion mechanism.

3.7. Fusion layer

The Fusion Layer aggregates the attended features $\tilde{\mathbf{Z}}$ into a compact and task-specific representation through a learnable weighted summation:

$$\mathbf{h}_{\text{fusion}} = \sum_{i=1}^{LR} w_i \tilde{\mathbf{Z}}_i \quad (18)$$

where $\mathbf{w} = (w_1, w_2, \dots, w_{LR})$ denotes a set of trainable scalar weights that determine the relative contribution of each layer-modality feature to the fused representation. The resulting fused feature vector $\mathbf{h}_{\text{fusion}} \in \mathbb{R}^d$ is then input to a multi-layer perceptron (MLP) for final property prediction:

$$\mathbf{y} = \text{MLP}(\mathbf{h}_{\text{fusion}}) \quad (19)$$

3.7.1. Dendritic neuron module (DNM)

We employ a *Dendritic Neuron Module* (DNM) to strengthen the expressiveness of handcrafted molecular fingerprints while retaining interpretability. Motivated by the compartmentalized computation of biological neurons, DNM encapsulates synaptic integration, dendritic modulation, and somatic aggregation in a single differentiable block, thereby enabling richer nonlinear transformations than a standard fully connected layer. Formally, given an input fingerprint vector $\mathbf{x} \in \mathbb{R}^d$ and an output dimension o , DNM computes

$$\mathbf{y} = \sum_{m=1}^M \sigma(\mathbf{x} \mathbf{W}_m + \mathbf{b}_m) \odot \tanh(\mathbf{m}_m) \in \mathbb{R}^o, \quad (20)$$

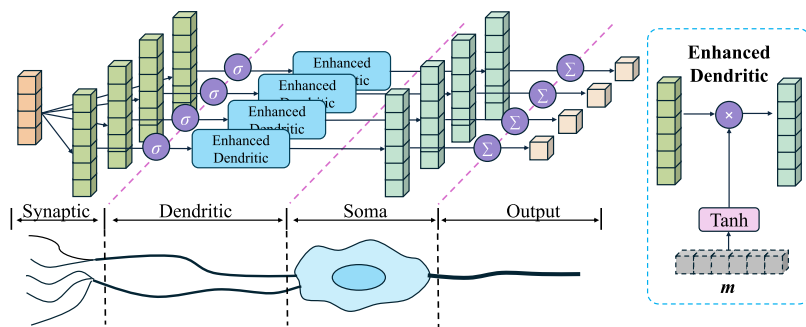


Fig. 4. Overall architecture of the DNM.

where M is the number of dendritic branches, $\sigma(\cdot)$ is the element-wise sigmoid for synaptic responses, $\tanh(\cdot)$ captures dendritic nonlinearity, $\mathbf{W}_m \in \mathbb{R}^{d \times o}$ and $\mathbf{b}_m \in \mathbb{R}^o$ are learnable synaptic parameters for branch m , and $\mathbf{m}_m \in \mathbb{R}^o$ are learnable dendritic modulation parameters. This branchwise decomposition explicitly separates *what* is transmitted (synaptic gating) from *how* it is emphasized or suppressed (dendritic modulation), before the soma aggregates the results. The overall architecture is illustrated in Fig. 4.

For clarity, the internal computation can be viewed through the following staged interpretation-without altering the above formulation. First, synaptic integration maps the input to M branchwise responses for each output unit:

$$\mathbf{S} = \sigma(\mathbf{x}\mathbf{W} + \mathbf{b}) \in \mathbb{R}^{o \times M}, \quad \mathbf{W} \in \mathbb{R}^{d \times o \times M}, \quad \mathbf{b} \in \mathbb{R}^{o \times M}. \quad (21)$$

Next, dendritic modulation provides per-branch nonlinear amplification or suppression via learnable parameters:

$$\mathbf{P} = \tanh(\mathbf{m}) \in \mathbb{R}^{o \times M}, \quad (22)$$

where the use of $\tanh(\cdot)$ nonlinearity for dendritic modulation is motivated by both biological plausibility and empirical stability. Unlike unbounded activations, $\tanh(\cdot)$ provides a symmetric and bounded response, allowing dendritic branches to both amplify and suppress synaptic inputs in a controlled manner [22]. The modulated dendritic responses are then obtained by element-wise combination:

$$\mathbf{D} = \mathbf{S} \odot \mathbf{P} \in \mathbb{R}^{o \times M}. \quad (23)$$

At the soma, responses are pooled across branches to produce a membrane-potential-like intermediate:

$$\mathbf{y}_j = \sum_{k=1}^M \mathbf{D}_{j,k}, \quad j = 1, \dots, o. \quad (24)$$

Conceptually, DNM replaces a single global transformation with a structured, branchwise computation that better reflects the hierarchical flow of information in real neurons: inputs undergo gated synaptic transmission, are selectively amplified or inhibited by dendritic subcompartments, and are then integrated at the soma. This factorization introduces meaningful inductive biases (competition, saturation, and branch specialization), improves parameter efficiency compared with stacking multiple dense layers, and promotes training stability through bounded activations.

Prior studies report that dendritic mechanisms and alternating excitation-inhibition can enhance robustness and generalization in classification [22], and dendritic-inspired modules have shown versatility across domains such as image recognition and medical image segmentation [32,33]. Molecular fingerprints are high-dimensional and sparse, where predictive patterns often rely on selective combinations of substructures rather than linear feature aggregation. The dendritic neuron module introduces multiplicative gating, enabling conjunction-like interactions among fingerprint features, which are difficult to model efficiently with standard MLPs. In our setting, placing DNM in the fingerprint branch offers a compact yet expressive transformation that is

well aligned with these findings, providing both performance gains and interpretability [34].

4. Experiments

4.1. Experimental settings

4.1.1. Datasets

To systematically assess the generalizability and robustness of our approach, we conducted experiments on a suite of benchmark datasets from MoleculeNet, a widely adopted resource for molecular property prediction. Specifically, our evaluation covers six classification datasets and four regression datasets, spanning a range of tasks including toxicity classification, blood-brain barrier penetration, solubility estimation, and quantum mechanical property prediction. All datasets were accessed via the DeepChem library [35], ensuring consistent preprocessing and standardized random data splits. We followed DeepChem's default partitioning protocols, dividing each dataset into training, validation, and test subsets to facilitate reproducibility and fair comparison. For classification tasks, we report results using the ROC-AUC(†) or PRC-AUC(†) metrics, depending on the task as suggested by MoleculeNet. For regression tasks, performance is evaluated using RMSE(†) or MAE(†). Detailed statistics for each dataset, including task type, evaluation metric, data split sizes, and prediction targets, are provided in Table 1.

4.1.2. Baselines

We compared our proposed method against a diverse set of baseline models representative of different molecular representation paradigms. For a fair and consistent comparison, the majority of baseline models were re-implemented or re-trained in our experimental environment using the same data splits, evaluation protocols, and performance metrics as MoLA. We followed the standard model architectures and hyperparameter settings reported in the original papers or official implementations whenever available. All experiments were conducted under the same training and evaluation settings to ensure reproducibility and fairness.

In addition, we include MolCL-SP [36] and MolMFD [37] as representative recent multimodal baselines for molecular property classification. Since the original papers only report classification results on MoleculeNet and do not provide complete experimental configurations or regression benchmarks, we directly quote their reported classification performance for reference. These models are therefore only compared on classification tasks, while regression comparisons are conducted using re-implemented baselines with available configurations.

- **Fingerprint-based models:** MLP, XGBoost, and IFM-MLP [7], which operate on handcrafted molecular fingerprints.
- **Graph-based models:** GCN [38], GAT [39], MPNN [40], and MHG-GNN [41], which learn molecular representations through message passing or hierarchical graph aggregation.

Table 1
Dataset statistics and target property descriptions for all benchmarks.

Dataset	Dataset Type	Tasks	Metric	Dataset Size	Target Prediction
BACE	Classification	1	ROC-AUC	{1210, 151, 152}	Inhibition of human beta-secretase 1
BBBP	Classification	1	ROC-AUC	{1631, 204, 204}	Blood-brain barrier penetration
ClinTox	Classification	2	ROC-AUC	{1184, 148, 148}	Clinical trial toxicity
MUV	Classification	17	PRC-AUC	{74469, 9309, 9309}	Biological assay active compound identification
SIDER	Classification	27	ROC-AUC	{1141, 143, 143}	Drug side-effect prediction
Tox21	Classification	12	ROC-AUC	{6258, 782, 783}	Perturbation of 12 toxicity-related pathways
Delaney	Regression	1	RMSE	{902, 113, 113}	Aqueous solubility
FreeSolv	Regression	1	RMSE	{513, 64, 65}	Hydration free energy
Lipophilicity	Regression	1	RMSE	{3360, 420, 420}	Octanol-water partition coefficient
QM7	Regression	1	MAE	{5470, 684, 684}	Quantum mechanical atomization energies
QM8	Regression	16	MAE	{17397, 2175, 2175}	Electronic excitation energies and oscillator strengths
QM9	Regression	12	MAE	{105576, 13197, 13197}	Molecular quantum chemical properties

Table 2
Hyperparameters used for MoLA experiments.

Hyperparameter	Value
Number of Epochs	300
Batch Size	256
Learning Rate (LR)	0.001
Dropout Rate	0.3
Encoder Depth (L)	4
Number of Dendrites (M)	20
Hidden Dimension	256
Attention Heads (H)	8
Optimizer	Adam
Weight Decay	1×10^{-5}
Early Stopping Patience	30

- **SMILES-based models:** LSTM [42], Transformer [43], and MoLFormer [44], a large-scale molecular language model pretrained on extensive chemical corpora to capture rich chemical semantics.
- **Hybrid/multimodal models:** FPGNN [12] and PremuNet-L [20], which combine fingerprint and graph-based representations as inputs. Multi-view [28] further combines MHGGNN with the frozen MoLFormer encoder to utilize both graph and language-based features. MolCL-SP [36] employs multimodal contrastive learning over substructure-level and global molecular representations for property classification. MolMFD [37] focuses on multimodal feature disentanglement and fusion across complementary molecular views to enhance predictive performance.

4.1.3. Experimental setup

All experiments were performed on a workstation running Windows 11 with an Intel Core i9-12900K CPU and an NVIDIA RTX 3090 GPU. All models were implemented in PyTorch 2.1 and Python 3.10, and hyperparameters were tuned uniformly across datasets. To control for randomness in initialization and data ordering, each method was executed in ten independent runs with different random seeds and results are reported as the mean and standard deviation. Complete implementation details and hyperparameter settings are provided in Table 2.

4.2. Results of classification

We comprehensively evaluate MoLA on six molecular classification datasets, including binary tasks (BACE, BBBP) and multi-label tasks (ClinTox, SIDER, MUV, Tox21). For evaluation, ROC-AUC is used on all datasets except MUV, where PRC-AUC is adopted to address its severe class imbalance. Table 3 reports the results, with model types categorized by their input representation. In the table, **bold** values denote the best performance for each dataset, and underlined values denote the second-best performance.

Across all datasets, MoLA consistently achieves strong performance, excelling in both binary and multi-label scenarios. Compared to traditional unimodal baselines, MoLA yields substantial improvements. Among multimodal baselines (FPGNN, PremuNet-L, Multi-view), MoLA also demonstrates clear superiority. Notably, although MultiView incorporates a pretrained MolFormer encoder, its performance is limited by parameter freezing, resulting in lower scores than MoLA and even MolFormer alone on some tasks.

MoLA achieves the best or second-best results on nearly all benchmarks, including top performance on BBBP, SIDER, MUV, and Tox21, and strong results on ClinTox and BACE. This demonstrates the effectiveness of MoLA's Cross-layer Attention and dendritic neuron-based fusion for capturing complementary information among diverse molecular modalities.

4.3. Results of regression

Table 4 presents the performance of various models on six molecular regression benchmarks. For Delaney, FreeSolv, and Lipophilicity, RMSE is reported, while for QM7, QM8, and QM9, MAE is used. MoLA consistently surpasses unimodal baselines and hybrid models, including FPGNN, PremuNet-L, and MultiView. Notably, MoLA achieves state-of-the-art or highly competitive results on Delaney, QM7, QM8, and QM9, indicating strong generalization and predictive performance.

These findings highlight the value of MoLA's adaptive multimodal fusion and dendritic neuron architecture, enabling effective modeling of complex, non-linear relationships in molecular property prediction tasks.

4.4. Visualization of multimodal fusion

To demonstrate the interpretability and rationality of MoLA's multimodal fusion strategy, we visualize the learned attention maps and fusion weights on the BBBP dataset, as shown in Fig. 5.

Fig. 5(A) shows the Cross-layer Attention Map, illustrating how MoLA dynamically allocates attention to Fingerprint (FP), SMILES, and Graph modalities across network layers. The results reveal that FP and SMILES representations receive higher attention scores, indicating their dominant roles in the fusion process. This aligns with Fig. 5(C), where models utilizing FP and SMILES achieve superior predictive performance compared to those using only Graph features. Fig. 5(B) displays the fusion weights learned at each layer. The Fingerprint modality consistently obtains larger weights, especially in early layers, while Graph features contribute less but provide complementary information when fused with FP and SMILES.

These visualizations confirm that MoLA adaptively and effectively integrates complementary information from multiple modalities, providing a clear and interpretable understanding of the model's fusion behavior.

Table 3

Classification results (ROC-AUC¹ / PRC-AUC²) of MoLA and baseline models on six MoleculeNet classification benchmarks.

Method	Data Type	BACE ¹	BBBP ¹	ClinTox ¹	MUV ²	SIDER ¹	Tox21 ¹	Avg
MLP	Fingerprint	83.0 ± 0.7	90.1 ± 0.3	58.1 ± 4.7	5.7 ± 0.7	62.0 ± 0.4	78.9 ± 0.1	62.96
XGBoost	Fingerprint	86.7 ± 0.0	93.0 ± 0.0	81.9 ± 0.0	4.8 ± 0.0	67.3 ± 0.0	80.8 ± 0.0	69.09
IFMMLP	Fingerprint	87.6 ± 3.4	91.4 ± 2.2	84.5 ± 4.1	9.3 ± 4.2	67.0 ± 2.4	79.2 ± 1.2	69.85
GCN	Graph	72.7 ± 6.1	83.1 ± 3.0	61.1 ± 15.1	0.4 ± 0.2	56.0 ± 4.0	76.6 ± 1.5	58.32
GAT	Graph	78.8 ± 0.8	84.1 ± 0.5	80.9 ± 1.9	8.3 ± 1.1	56.1 ± 0.5	80.7 ± 0.4	64.83
MPNN	Graph	83.4 ± 4.1	88.6 ± 4.6	87.0 ± 16.0	4.1 ± 3.2	60.7 ± 2.5	79.2 ± 1.6	67.14
MHGGNN	Graph	84.8 ± 2.7	88.2 ± 3.4	83.9 ± 8.1	2.5 ± 1.3	61.5 ± 3.0	80.7 ± 1.9	66.92
LSTM	SMILES	81.3 ± 8.2	89.6 ± 1.7	69.0 ± 20.1	0.1 ± 0.1	51.6 ± 1.8	69.9 ± 9.8	60.26
Transformer	SMILES	79.4 ± 3.2	91.6 ± 2.2	94.6 ± 2.4	0.2 ± 0.2	58.5 ± 2.2	73.7 ± 1.8	66.35
MoLFormer	SMILES	81.1 ± 0.4	94.4 ± 0.2	98.2 ± 0.1	10.0 ± 0.6	67.0 ± 1.3	82.9 ± 0.2	72.25
FPGNN	Hybrid	82.7 ± 0.9	93.1 ± 0.3	53.9 ± 2.1	5.9 ± 1.8	64.5 ± 0.4	79.0 ± 0.4	63.19
PremuNet-L	Hybrid	87.9 ± 2.4	89.6 ± 3.1	70.8 ± 10.3	2.8 ± 2.2	61.0 ± 1.9	75.0 ± 1.0	64.54
Multi-view	Hybrid	85.4 ± 2.0	88.1 ± 1.1	85.0 ± 3.4	1.6 ± 0.8	62.3 ± 2.8	80.2 ± 0.6	67.09
MolCL-SP	Hybrid	82.9 ± 1.0	74.4 ± 0.7	99.4 ± 0.2	–	65.6 ± 0.1	76.9 ± 1.0	–
MolMFD	Hybrid	86.9 ± 0.5	74.8 ± 1.5	92.3 ± 0.8	–	66.1 ± 0.9	80.0 ± 0.3	–
MoLA (Ours)	Hybrid	87.5 ± 0.4	95.3 ± 0.4	97.0 ± 0.3	11.8 ± 1.5	69.1 ± 0.4	84.1 ± 0.3	74.14

Table 4

Regression results (RMSE¹ / MAE²) of MoLA and baseline models on six MoleculeNet regression benchmarks.

Method	Data Type	Delaney ¹	FreeSolv ¹	Lipophilicity ¹	QM7 ²	QM8 ²	QM9 ²	Avg
MLP	Fingerprint	0.660 ± 0.011	0.427 ± 0.007	0.669 ± 0.005	0.778 ± 0.002	0.671 ± 0.002	0.737 ± 0.001	0.657
XGBoost	Fingerprint	0.587 ± 0.000	0.325 ± 0.000	0.687 ± 0.000	0.787 ± 0.000	0.655 ± 0.000	0.742 ± 0.000	0.630
IFMMLP	Fingerprint	0.808 ± 0.165	0.864 ± 0.029	0.730 ± 0.094	0.778 ± 0.013	0.675 ± 0.032	0.766 ± 0.003	0.770
GCN	Graph	0.617 ± 0.006	0.312 ± 0.009	0.830 ± 0.012	0.784 ± 0.001	0.684 ± 0.002	0.763 ± 0.000	0.665
GAT	Graph	0.486 ± 0.009	0.293 ± 0.005	0.798 ± 0.047	0.784 ± 0.002	0.674 ± 0.004	0.759 ± 0.000	0.632
MPNN	Graph	0.478 ± 0.018	0.253 ± 0.007	0.716 ± 0.006	0.780 ± 0.002	0.667 ± 0.007	0.756 ± 0.001	0.609
MHGGNN	Graph	0.595 ± 0.084	0.598 ± 0.103	0.629 ± 0.026	0.770 ± 0.010	0.668 ± 0.003	0.741 ± 0.001	0.667
LSTM	SMILES	1.035 ± 0.002	0.877 ± 0.002	1.010 ± 0.001	0.814 ± 0.000	0.678 ± 0.005	0.749 ± 0.002	0.861
Transformer	SMILES	1.046 ± 0.020	0.889 ± 0.012	1.008 ± 0.006	0.813 ± 0.001	0.668 ± 0.005	0.763 ± 0.000	0.865
MoLFormer	SMILES	0.349 ± 0.008	0.333 ± 0.007	0.666 ± 0.015	0.735 ± 0.005	0.661 ± 0.002	0.748 ± 0.000	0.582
FPGNN	Hybrid	0.573 ± 0.015	0.341 ± 0.008	0.659 ± 0.007	0.777 ± 0.003	0.667 ± 0.003	0.737 ± 0.001	0.626
PremuNet-L	Hybrid	0.470 ± 0.007	0.245 ± 0.013	0.636 ± 0.007	0.779 ± 0.005	0.666 ± 0.003	0.749 ± 0.001	0.591
Multi-view	Hybrid	0.688 ± 0.030	0.391 ± 0.019	0.802 ± 0.017	0.819 ± 0.005	0.689 ± 0.004	0.762 ± 0.001	0.692
MoLA (Ours)	Hybrid	0.347 ± 0.017	0.271 ± 0.026	0.636 ± 0.009	0.721 ± 0.011	0.635 ± 0.006	0.729 ± 0.002	0.556

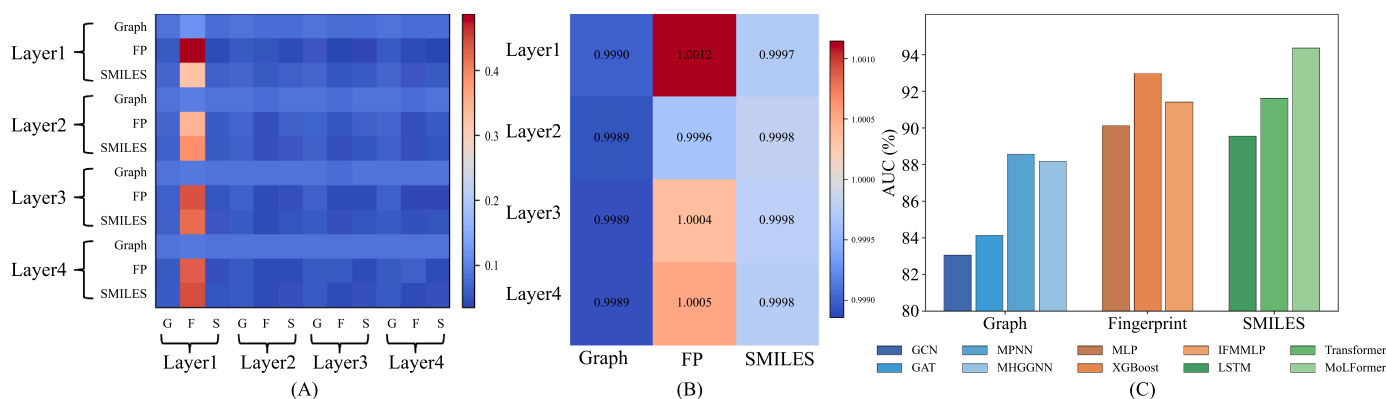


Fig. 5. Visualization of MoLA's multimodal fusion mechanisms on the BBBP dataset. (A) Cross-layer Attention Map showing the allocation of attention to Graph, Fingerprint (FP), and SMILES modalities across layers. (B) Fusion Layer weights for each modality at different network depths. (C) Model performance comparison using different molecular representations, evaluated by AUC scores.

4.5. Ablation study

This section focuses on ablation experiments for the key architectural components and input modalities of MoLA.

To assess the contribution of core architectural components and molecular modalities in MoLA, we conduct a comprehensive set of ablation experiments, as summarized in Table 5. Structural ablations are

summarized as follows: (1) Removing the Concat Layer (**w/o CL**), which aggregates hidden states across encoder layers, leads to a pronounced drop in performance, demonstrating the necessity of hierarchical feature aggregation. (2) Excluding the Cross-Layer Attention Layer (**w/o CA**), which adaptively integrates information from all layers and modalities, results in further performance decline, indicating the critical role of adaptive cross-layer information flow. (3) Eliminating the Fusion Layer

Table 5
Impact of core architectural modules on MoLA’s performance.

Dataset	MoLA	w/o CL	w/o CA	w/o FL	w/o CA&FL
BACE	87.5 ± 0.4	86.3 ± 0.8	87.2 ± 0.5	<u>87.4 ± 0.3</u>	87.4 ± 0.4
BBBP	95.3 ± 0.4	95.1 ± 0.2	95.1 ± 0.2	<u>95.2 ± 0.3</u>	95.1 ± 0.2
ClinTox	<u>97.0 ± 0.3</u>	97.2 ± 0.3	96.8 ± 0.3	<u>96.6 ± 0.5</u>	96.7 ± 0.3
MUV	11.8 ± 1.5	3.3 ± 1.1	8.1 ± 2.0	<u>10.5 ± 3.2</u>	8.9 ± 2.5
SIDER	69.1 ± 0.4	66.0 ± 0.4	68.0 ± 0.6	<u>69.1 ± 0.3</u>	68.7 ± 0.5
Tox21	84.1 ± 0.3	82.8 ± 0.2	84.3 ± 0.2	84.1 ± 0.2	<u>84.2 ± 0.2</u>
Avg	74.14	71.77	73.24	<u>73.82</u>	73.51

Table 6

Ablation study on modality contributions. The fingerprint (FP) branch in MoLA consists of the concatenation of traditional molecular fingerprints and pretrained MolFormer embeddings. “w/o FP” removes the entire fingerprint branch, while “w/o MolFormer” removes only the MolFormer component and retains traditional fingerprints.

Dataset	MoLA	w/o FP	w/o MolFormer	w/o Graph	w/o SMILES
BACE	87.5 ± 0.4	85.9 ± 1.4	91.0 ± 1.8	87.5 ± 0.3	86.9 ± 1.0
BBBP	<u>95.3 ± 0.4</u>	95.9 ± 0.3	94.2 ± 1.8	95.1 ± 0.4	94.9 ± 0.2
ClinTox	<u>97.0 ± 0.3</u>	95.2 ± 1.6	92.7 ± 5.4	97.2 ± 0.4	96.2 ± 1.3
MUV	11.8 ± 1.5	7.0 ± 1.4	6.5 ± 2.0	<u>11.3 ± 1.9</u>	9.8 ± 2.6
SIDER	69.1 ± 0.4	60.8 ± 0.5	66.0 ± 1.2	<u>68.7 ± 0.4</u>	<u>68.8 ± 0.3</u>
Tox21	<u>84.1 ± 0.3</u>	83.0 ± 0.7	79.9 ± 2.3	84.2 ± 0.3	84.1 ± 0.2
Avg	74.14	71.32	71.70	<u>73.99</u>	73.47

(w/o FL), which synthesizes multimodal features for final prediction, also causes a slight decrease in accuracy. (4) When both the Cross-Layer Attention and Fusion Layer are ablated simultaneously (w/o CA&FL), the model’s performance is lower than the full model and the w/o FL variant, but remains higher than w/o CA. This result indicates that, even without both modules, the model still benefits from basic multimodal aggregation, but lacks the adaptive integration and deep fusion provided by the dedicated layers, leading to intermediate performance.

For modality ablations, we further assess the contribution of each input type by selectively removing individual molecular representations: (1) Excluding the entire fingerprint branch leads to substantial performance declines, particularly on chemically diverse or challenging datasets such as SIDER and MUV, underlining the critical role of fingerprint-based information in capturing molecular characteristics. (2) Removing only the MolFormer embeddings while retaining traditional fingerprints also results in a clear performance drop, indicating that the pretrained MolFormer representations provide complementary information beyond handcrafted fingerprints, but are not solely responsible for the overall performance gains. (3) Removing the graph modality results in a smaller performance decrease, suggesting partial redundancy for certain tasks where structural information can be compensated by other modalities. (4) Omitting SMILES features yields an intermediate drop, especially on endpoints that benefit from sequence-level information.

These results, as reported in Tables 5 and 6, collectively demonstrate that both the hierarchical architectural modules and comprehensive multimodal representations are indispensable for achieving state-of-the-art molecular property prediction with MoLA. The ablation studies provide strong empirical evidence for the rationality and effectiveness of our model design.

4.6. Comparison of dendritic neuron module and MLP

To assess the effectiveness of the dendritic neuron module (DNM), we conducted comparative experiments with the traditional multi-layer perceptron (MLP) across several molecular property classification benchmarks. As presented in Fig. 6, DNM consistently achieves higher performance than MLP, underscoring its enhanced capacity to model complex and nonlinear feature interactions in molecular data.

We also evaluated the influence of the number of dendrites (M) in DNM. The results demonstrate that increasing M generally leads to improved prediction accuracy, highlighting the benefit of richer dendritic integration. However, the marginal gain diminishes when M exceeds 20, suggesting an optimal range for dendritic complexity that balances expressivity and efficiency.

These findings indicate that the dendritic neuron architecture not only provides superior predictive power compared to standard MLPs but also exhibits robustness with respect to architectural hyperparameters, making it a promising component for molecular representation learning.

4.7. Impact of encoder depth

To evaluate the impact of encoder depth (L) in MoLA, we conduct a systematic study with $L \in \{1, 2, 3, 4, 6, 8\}$ layers, assessing model performance across multiple molecular property prediction datasets. Deeper layers enable broader receptive fields and more expressive multimodal interactions, but may also introduce redundancy or over-smoothing.

Fig. 7 presents box plots summarizing results over 10 independent runs. As shown, most datasets benefit from increasing L up to 4, beyond which performance often plateaus or declines. A few datasets (e.g., BACE, SIDER, Tox21) continue to improve with deeper encoders, highlighting some variability across tasks.

Table 7 reports the average AUC across all datasets. Notably, $L = 4$ achieves the highest overall performance, outperforming both shallower and deeper configurations. Considering its superior average accuracy and lower computational cost, we adopt $L = 4$ as the default setting for all main experiments.

4.8. Impact of encoder architecture across modalities

To assess how different encoder architectures affect model performance in multimodal molecular learning, we conduct a systematic comparison of various encoder configurations for each input modality: fingerprint ($\mathcal{E}_F$), SMILES ($\mathcal{E}_S$), and molecular graph ($\mathcal{E}_G$). Specifically, we evaluate four combinations covering representative architectures: Dendritic Neuron Module (DNM) and MLP for fingerprints, Transformer and LSTM for SMILES, and GIN and GAT for graphs.

Fingerprint Encoder ($\mathcal{E}_F$). Comparing the first and last rows of Table 8, replacing MLP with DNM for fingerprint encoding leads to a noticeable improvement in average performance (from 72.24% to 74.14%). This demonstrates the benefit of adopting a more structured, biologically inspired transformation for dense fingerprint vectors. Notably, on the challenging MUV dataset, this replacement leads to a gain of over 8 percentage points (from 3.3% to 11.8%).

SMILES Encoder ($\mathcal{E}_S$). We explore both Transformer and LSTM architectures for SMILES sequences. While both show competitive results, Transformer-based encoders generally outperform LSTM counterparts, especially when paired with strong fingerprint and graph encoders. For example, under the configuration DNM-Transformer-GIN, the model achieves the highest overall average, while replacing Transformer with LSTM slightly reduces performance. This indicates that the self-attention mechanism in Transformers may better capture long-range token dependencies in SMILES strings.

Graph Encoder ($\mathcal{E}_G$). For molecular graphs, we compare GIN and GAT as structural encoders. GIN generally performs better across most tasks. The best average score and strongest results on datasets like MUV, SIDER, and Tox21 are achieved when GIN is used. For example, in the DNM-Transformer-GIN setup, GIN leads to stronger overall performance than GAT under DNM-Transformer-GAT. This aligns with prior findings suggesting that GIN’s injective neighborhood aggregation is well-suited for molecular graph learning [45].

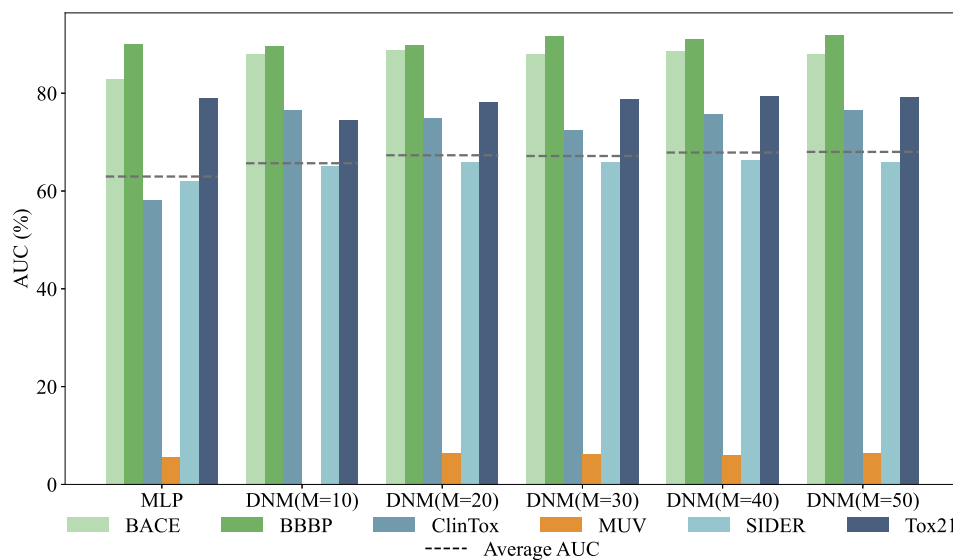


Fig. 6. Comparison of Dendritic Neuron Module (DNM) and traditional MLP on molecular property prediction. The figure also shows the effect of varying the number of dendrites (M) in DNM.

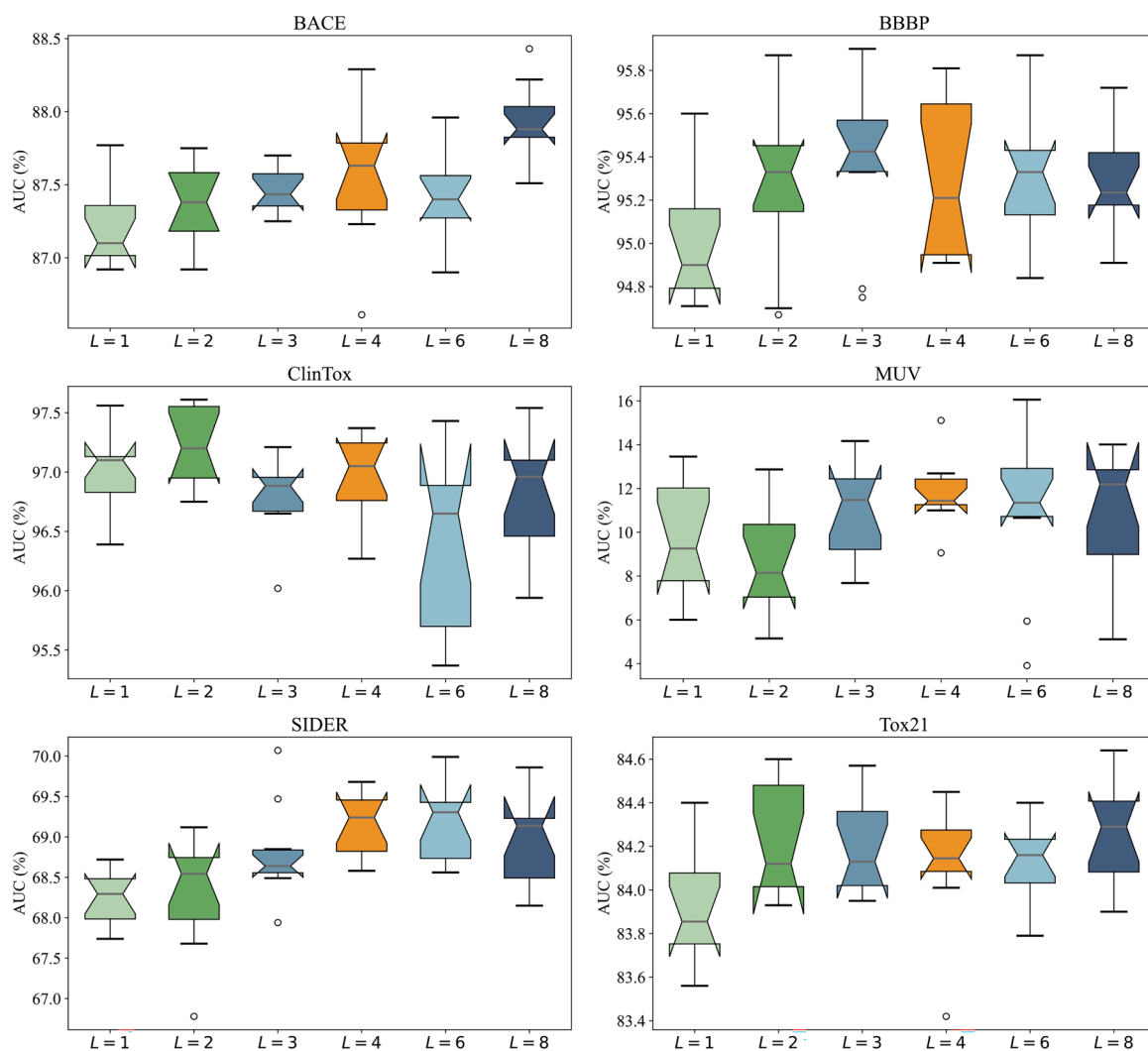


Fig. 7. Effect of encoder depth (L) on the distribution of test set AUC for the classification datasets.

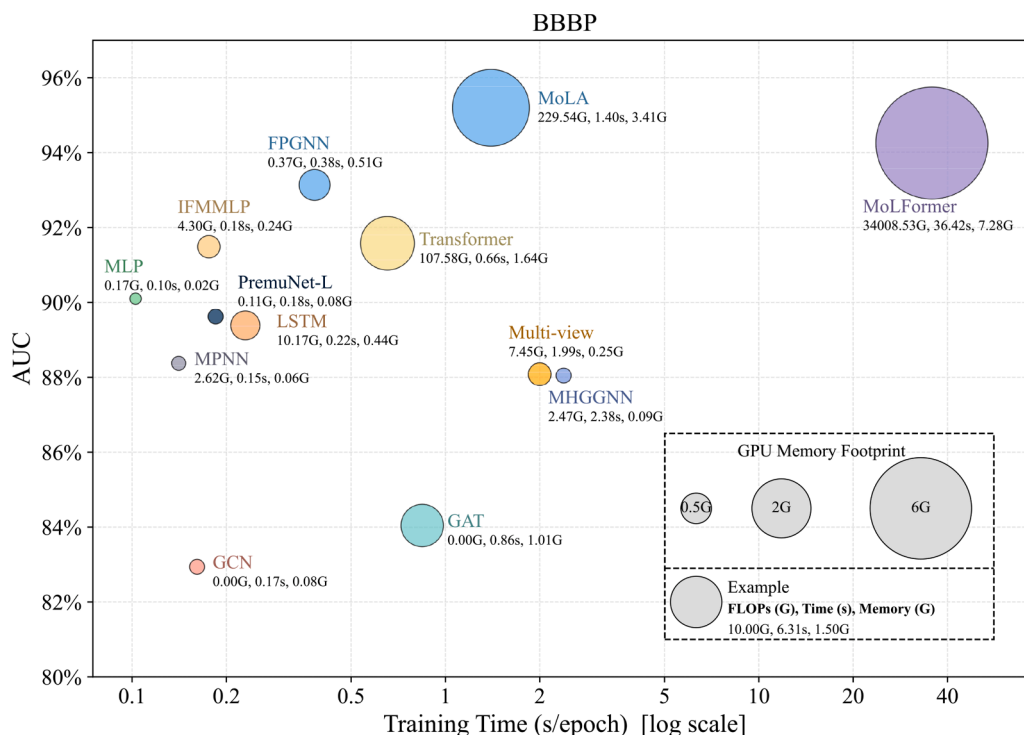


Fig. 8. Performance-efficiency trade-off on the BBBP benchmark, comparing AUC, training time per epoch, FLOPs, and GPU memory usage. Bubble size denotes GPU memory consumption.

Table 7

Effect of encoder depth (L) on the performance of MoLA across different benchmark datasets.

Method	BACE	BBBP	ClinTox	MUV	SIDER	Tox21	Avg
$L = 1$	87.2 ± 0.3	95.0 ± 0.3	97.0 ± 0.3	9.7 ± 2.6	68.3 ± 0.3	83.9 ± 0.2	73.51
$L = 2$	87.4 ± 0.3	95.3 ± 0.4	97.2 ± 0.3	8.6 ± 2.3	68.3 ± 0.7	84.2 ± 0.3	73.50
$L = 3$	87.5 ± 0.1	95.4 ± 0.4	96.8 ± 0.3	11.1 ± 2.2	68.8 ± 0.6	84.2 ± 0.2	73.96
$L = 4$	87.5 ± 0.4	95.3 ± 0.4	97.0 ± 0.3	11.8 ± 1.5	69.1 ± 0.4	84.1 ± 0.3	74.14
$L = 6$	87.4 ± 0.3	95.3 ± 0.3	96.4 ± 0.7	11.0 ± 3.4	69.2 ± 0.4	84.1 ± 0.2	73.90
$L = 8$	87.9 ± 0.3	95.3 ± 0.3	96.8 ± 0.5	10.9 ± 2.8	69.0 ± 0.5	84.3 ± 0.2	74.02

Table 8

Effect of encoder architecture choices for fingerprints, SMILES, and graphs on MoLA performance across datasets.

$\mathcal{E}_F$	$\mathcal{E}_S$	$\mathcal{E}_G$	BACE	BBBP	ClinTox	MUV	SIDER	Tox21	Avg
MLP	Transformer	GIN	87.7 $\pm$ 0.6	95.2 ± 0.2	94.7 ± 1.2	3.3 ± 1.2	68.9 ± 1.3	83.5 ± 0.3	72.24
DNM	Transformer	GAT	87.4 ± 0.3	<u>95.3 ± 0.3</u>	96.9 ± 0.5	10.1 ± 2.1	68.8 ± 0.6	84.1 $\pm$ 0.2	73.79
DNM	LSTM	GIN	86.9 ± 0.4	95.2 ± 0.2	97.5 $\pm$ 0.2	<u>10.9 ± 2.6</u>	<u>69.1 ± 0.4</u>	84.0 ± 0.3	<u>73.95</u>
DNM	Transformer	GIN	<u>87.5 ± 0.4</u>	95.3 $\pm$ 0.4	<u>97.0 ± 0.3</u>	11.8 $\pm$ 1.5	69.1 $\pm$ 0.4	<u>84.1 ± 0.3</u>	74.14

Cross-modal Synergy. The best-performing configuration (DNM-Transformer-GIN) consistently achieves top-2 results across all benchmarks, highlighting the importance of modality-specific architecture selection. The observed performance gains are not attributable to a single encoder but rather emerge from the synergistic combination of encoders suited to the nature of each modality.

Outlook. These results underscore the modular and flexible nature of the MoLA framework, which allows for systematic replacement or upgrading of encoders without altering the overall architecture. As more powerful encoder backbones (e.g., graph transformers, pretrained SMILES models, or improved neuron-inspired encoders) become available, MoLA is well-positioned to integrate them and further enhance performance. This adaptability makes MoLA a promising foundation for future research in multimodal molecular representation learning.

4.9. Robustness to missing modalities

To simulate realistic scenarios where some molecular modalities might be unavailable at inference time, we conduct a missing-modality evaluation by testing the trained MoLA model under different combinations of absent inputs. Table 9 reports the performance when one or more modalities are deliberately removed during inference. All models are trained with the complete set of modalities (traditional fingerprints, pretrained MoLFormer embeddings, graph, and SMILES), and only the inference-time input is altered by zeroing out the corresponding modality embeddings.

Note that in this section we treat *traditional fingerprints* and *MoLFormer embeddings* as two separate modalities: “Fingerprint” refers to the handcrafted fingerprint features only, while “MoLFormer” denotes the pretrained MoLFormer representation. Therefore, “w/o Fingerprint” removes only the traditional fingerprint input, whereas “w/o MoLFormer”

Table 9

Performance of the trained MoLA model under different missing modality configurations at inference time.

Fingerprint	MolFormer	Graph	SMILES	BACE	BBBP	ClinTox	MUV	SIDER	Tox21	Avg
✓	✓	✓		86.3 ± 0.5	94.8 ± 0.3	95.0 ± 0.9	11.8 ± 1.5	69.1 ± 0.4	84.0 ± 0.3	73.50
✓	✓		✓	87.1 ± 0.9	94.2 ± 0.7	96.8 ± 0.5	11.9 ± 1.7	69.1 ± 0.4	84.1 ± 0.3	73.85
✓		✓	✓	87.1 ± 0.3	89.9 ± 0.7	89.4 ± 8.7	4.1 ± 0.6	64.6 ± 0.6	76.8 ± 0.5	68.65
	✓	✓	✓	76.0 ± 0.7	93.5 ± 0.8	98.0 ± 0.2	8.0 ± 1.9	66.5 ± 0.5	79.6 ± 0.3	70.26
		✓	✓	62.0 ± 0.2	77.4 ± 1.8	93.3 ± 0.8	0.1 ± 0.1	56.5 ± 1.6	54.6 ± 5.1	57.32
✓	✓			86.0 ± 0.8	93.6 ± 0.7	94.8 ± 0.6	11.9 ± 1.8	69.0 ± 0.4	84.0 ± 0.3	73.21
✓	✓	✓	✓	87.5 ± 0.4	95.3 ± 0.4	97.0 ± 0.3	11.8 ± 1.5	69.1 ± 0.4	84.1 ± 0.3	74.14

removes only the pretrained MolFormer input, while all other modalities remain unchanged.

These results demonstrate that MoLA retains competitive performance even when one or more input modalities are absent during inference. For instance, removing the graph modality or SMILES strings leads to only minor performance degradation, while removing the fingerprint or MolFormer modality causes a more noticeable drop, indicating their stronger contribution to the overall prediction. This robustness highlights MoLA's capacity to generalize across partial modality inputs, making it suitable for deployment in practical scenarios where certain molecular representations may be missing or incomplete.

4.10. Computational efficiency and resource analysis

We further evaluate MoLA in terms of model complexity and computational cost. Fig. 8 summarizes the predictive performance, training time, FLOPs, and GPU memory usage on the BBBP benchmark. The x-axis shows the wall-clock training time per epoch (log scale), the y-axis reports AUC, and the bubble size represents GPU memory consumption.

Despite integrating multiple lightweight encoders and a cross-layer attention mechanism, MoLA maintains moderate computational overhead in practice. Similar to the Multi-view baseline, MoLA uses pretrained MolFormer embeddings as fixed input features and does not require fine-tuning the MolFormer encoder. Consequently, MoLA incurs substantially lower FLOPs and GPU memory usage than the full MolFormer model, which is trained end-to-end.

Overall, MoLA achieves a favorable balance between performance and efficiency: it attains near-top AUC while operating in a significantly lower training-time and memory regime compared to MolFormer, and the additional cost over single-modality baselines remains controlled.

5. Conclusion

In this work, we present MoLA, a multimodal neural network for molecular property prediction that adaptively integrates fingerprint, SMILES, and graph-based representations through a unified cross-layer attention and fusion framework. By incorporating a dendritic neuron module and adaptive aggregation strategies, MoLA effectively captures complementary information across heterogeneous molecular modalities. Extensive experiments demonstrate state-of-the-art performance on multiple benchmarks, along with improved interpretability enabled by attention-based visualization. Ablation studies further validate the necessity of each component and highlight the benefits of coordinated multimodal integration.

We note that the current evaluation follows standard MoleculeNet protocols and focuses on clean, canonical molecular representations. Robustness under noisy or uncertain molecular inputs, such as perturbed SMILES strings, corrupted graph structures, or conformer variability, remains an important and challenging direction for future research [46]. More broadly, recent studies in related domains have explored robustness-aware learning under noisy or low-quality inputs, and extending MoLA to explicitly account for such uncertainty, potentially in combination with noise-aware training or uncertainty modeling techniques, represents a promising avenue for further investigation [47–49]. As molecular encoder architectures continue to advance, we believe

MoLA provides a flexible foundation for integrating richer representations while maintaining robustness and interpretability.

CRedit authorship contribution statement

Jiayi Li: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization; **Zihang Zhang:** Visualization, Validation, Software, Methodology, Formal analysis; **Zhenyu Lei:** Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Jiujun Cheng:** Supervision, Resources, Methodology, Formal analysis; **Lianbo Ma:** Supervision, Methodology, Investigation, Formal analysis; **Cong Liu:** Supervision, Investigation, Formal analysis; **Shangce Gao:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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